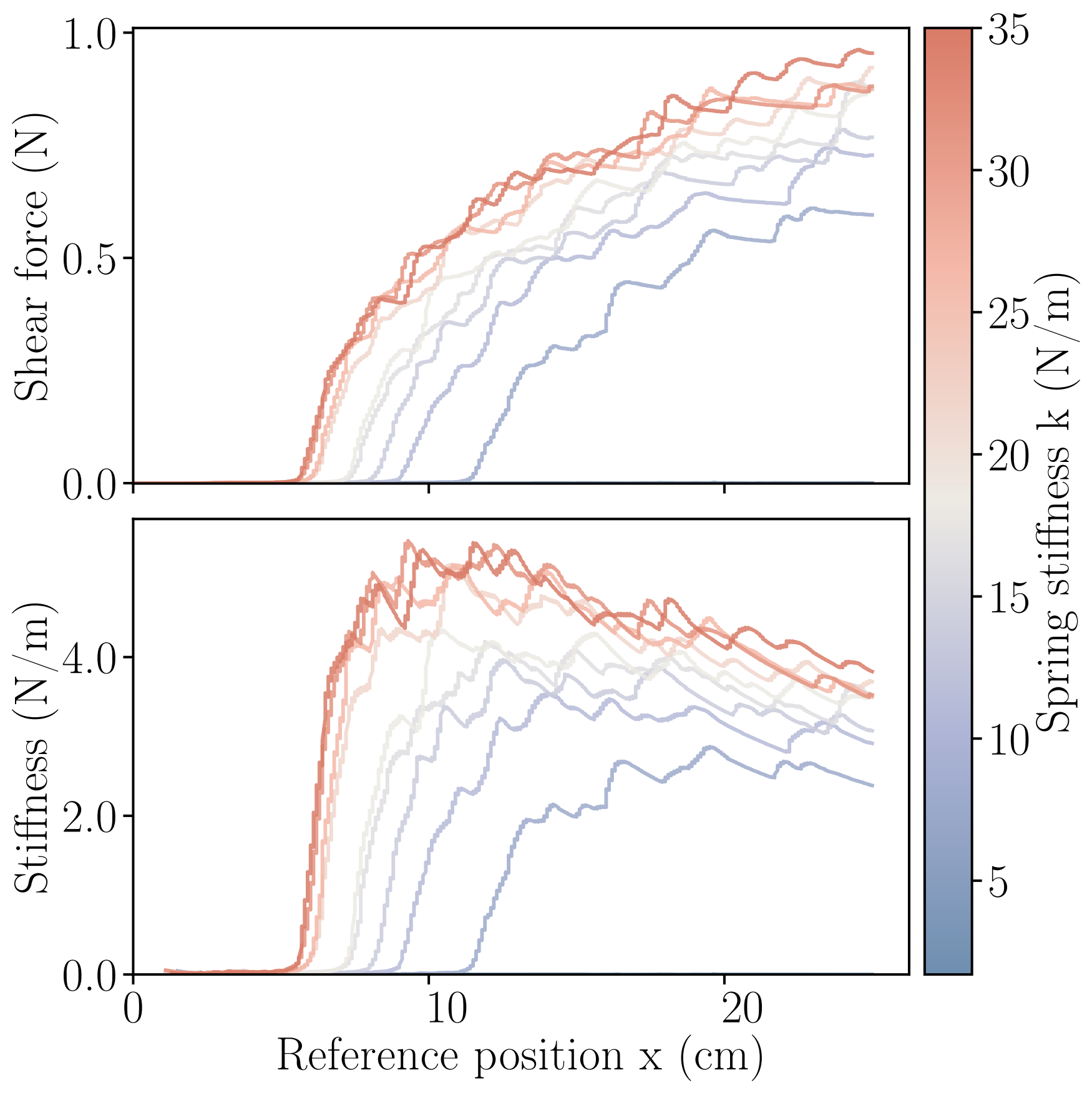

Shear force (N)
1.0
0.5
0.0
Stiffness (N/m)
4.0
2.0
0.0
0
10
20
Reference position x (cm)
Spring stiffness k (N/m)
35
30
25
20
15
10
5